WORLD HAPPENCESS DATA ANALYSIS

PROBLEM STATEMENT

One important research problem is to examine the key determinants of happiness across countries and to understand how economic, social, and institutional factors contribute to variations in happiness scores. Despite global development, countries show significant differences in happiness levels, and it remains unclear which factors play the most influential role in shaping these outcomes. Another research problem focuses on the relationship between income and happiness, specifically analyzing whether higher GDP per capita always leads to higher happiness levels. While economic theory often assumes a positive relationship, real-world data suggests that beyond a certain point, increases in income may have diminishing effects on happiness.

DAY 1

```
import pandas as pd
df = pd.read_csv("world-happiness-report-2021 - world-happiness-report-2021.csv")
df
```

	Country name	Regional indicator	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	P
0	Finland	Western Europe	7.842	0.032	7.904	7.780	10.775	0.954	72.000	0.949	-0.098	
1	Denmark	Western Europe	7.620	0.035	7.687	7.552	10.933	0.954	72.700	0.946	0.030	
2	Switzerland	Western Europe	7.571	0.036	7.643	7.500	11.117	0.942	74.400	0.919	0.025	
3	Iceland	Western Europe	7.554	0.059	7.670	7.438	10.878	0.983	73.000	0.955	0.160	
4	Netherlands	Western Europe	7.464	0.027	7.518	7.410	10.932	0.942	72.400	0.913	0.175	
...	...	...	...	...	...	...	...	...	...	...	...	
144	Lesotho	Sub-Saharan Africa	3.512	0.120	3.748	3.276	7.926	0.787	48.700	0.715	-0.131	
145	Botswana	Sub-Saharan Africa	3.467	0.074	3.611	3.322	9.782	0.784	59.269	0.824	-0.246	
146	Rwanda	Sub-Saharan Africa	3.415	0.068	3.548	3.282	7.676	0.552	61.400	0.897	0.061	
147	Zimbabwe	Sub-Saharan Africa	3.145	0.058	3.259	3.030	7.943	0.750	56.201	0.677	-0.047	
148	Afghanistan	South Asia	2.523	0.038	2.596	2.449	7.695	0.463	52.493	0.382	-0.102	

149 rows × 20 columns

```
df['Country name']
```

```
0          Finland
1          Denmark
2      Switzerland
3          Iceland
4      Netherlands
          ...
144        Lesotho
145       Botswana
146         Rwanda
147       Zimbabwe
148    Afghanistan
Name: Country name, Length: 149, dtype: str
```

```
df.columns
```

```
Index(['Country name', 'Regional indicator', 'Ladder score',
       'Standard error of ladder score', 'upperwhisker', 'lowerwhisker',
       'Logged GDP per capita', 'Social support', 'Healthy life expectancy',
       'Freedom to make life choices', 'Generosity',
       'Perceptions of corruption', 'Ladder score in Dystopia',
       'Explained by: Log GDP per capita', 'Explained by: Social support',
       'Explained by: Healthy life expectancy',
       'Explained by: Freedom to make life choices',
       'Explained by: Generosity', 'Explained by: Perceptions of corruption',
       'Dystopia + residual'],
      dtype='str')
```

```
df[["Country name", 'Regional indicator']]
```

	Country name	Regional indicator
0	Finland	Western Europe
1	Denmark	Western Europe
2	Switzerland	Western Europe
3	Iceland	Western Europe
4	Netherlands	Western Europe
...	...	...
144	Lesotho	Sub-Saharan Africa
145	Botswana	Sub-Saharan Africa
146	Rwanda	Sub-Saharan Africa
147	Zimbabwe	Sub-Saharan Africa
148	Afghanistan	South Asia

149 rows × 2 columns

```
df[(df['Country name']=='finland')]
```

Country name	Regional indicator	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Percept corrup

```
# selecting column
df['Country name']
```

```
0          Finland
1          Denmark
2      Switzerland
3          Iceland
4      Netherlands
          ...
144        Lesotho
145       Botswana
146         Rwanda
147       Zimbabwe
148    Afghanistan
Name: Country name, Length: 149, dtype: str
```

```
df.iloc[9] # single row # selecting row

df.iloc[4:11] # portion +
```

	Country name	Regional indicator	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Pe
4	Netherlands	Western Europe	7.464	0.027	7.518	7.410	10.932	0.942	72.4	0.913	0.175	
5	Norway	Western Europe	7.392	0.035	7.462	7.323	11.053	0.954	73.3	0.960	0.093	
6	Sweden	Western Europe	7.363	0.036	7.433	7.293	10.867	0.934	72.7	0.945	0.086	
7	Luxembourg	Western Europe	7.324	0.037	7.396	7.252	11.647	0.908	72.6	0.907	-0.034	
8	New Zealand	North America and ANZ	7.277	0.040	7.355	7.198	10.643	0.948	73.4	0.929	0.134	
9	Austria	Western Europe	7.268	0.036	7.337	7.198	10.906	0.934	73.3	0.908	0.042	
10	Australia	North America and ANZ	7.183	0.041	7.265	7.102	10.796	0.940	73.9	0.914	0.159	

```
df.rename(columns={"Country name":"country","Regional indicator":"Regional"},inplace=True)
```

```
print= df
```

```
print=df.iloc[9] # single row
```

INSPECT THE DATA

```
df.head
```

```
<bound method NDFrame.head of          country            Regional  Ladder score  \
0        Finland      Western Europe         7.842
1        Denmark      Western Europe         7.620
2    Switzerland      Western Europe         7.571
3        Iceland      Western Europe         7.554
4    Netherlands      Western Europe         7.464
..           ...                 ...           ...
144      Lesotho  Sub-Saharan Africa         3.512
145     Botswana  Sub-Saharan Africa         3.467
146       Rwanda  Sub-Saharan Africa         3.415
147     Zimbabwe  Sub-Saharan Africa         3.145
148  Afghanistan          South Asia         2.523

     Standard error of ladder score  upperwhisker  lowerwhisker  \
0                             0.032         7.904         7.780
1                             0.035         7.687         7.552
2                             0.036         7.643         7.500
3                             0.059         7.670         7.438
4                             0.027         7.518         7.410
..                              ...           ...           ...
144                           0.120         3.748         3.276
145                           0.074         3.611         3.322
146                           0.068         3.548         3.282
147                           0.058         3.259         3.030
148                           0.038         2.596         2.449

     Logged GDP per capita  Social support  Healthy life expectancy  \
0                   10.775           0.954                   72.000
1                   10.933           0.954                   72.700
2                   11.117           0.942                   74.400
3                   10.878           0.983                   73.000
4                   10.932           0.942                   72.400
..                     ...             ...                      ...
144                  7.926           0.787                   48.700
145                  9.782           0.784                   59.269
146                  7.676           0.552                   61.400
147                  7.943           0.750                   56.201
148                  7.695           0.463                   52.493

     Freedom to make life choices  Generosity  Perceptions of corruption  \
0                           0.949      -0.098                      0.186
1                           0.946       0.030                      0.179
2                           0.919       0.025                      0.292
3                           0.955       0.160                      0.673
4                           0.913       0.175                      0.338
```

```
..                          ...         ...                       ...
144                       0.715      -0.131                     0.915
145                       0.824      -0.246                     0.801
146                       0.897       0.061                     0.167
147                       0.677      -0.047                     0.821
148                       0.382      -0.102                     0.924

     Ladder score in Dystopia  Explained by: Log GDP per capita  \
0                        2.43                             1.446
1                        2.43                             1.502
2                        2.43                             1.566
3                        2.43                             1.482
4                        2.43                             1.501
```

```
df.tail
```

```
145                       0.824      -0.246                     0.801
146                       0.897       0.061                     0.167
147                       0.677      -0.047                     0.821
148                       0.382      -0.102                     0.924

     Ladder score in Dystopia  Explained by: Log GDP per capita  \
0                        2.43                             1.446
1                        2.43                             1.502
2                        2.43                             1.566
3                        2.43                             1.482
4                        2.43                             1.501
..                        ...                               ...
144                      2.43                             0.451
145                      2.43                             1.099
146                      2.43                             0.364
147                      2.43                             0.457
148                      2.43                             0.370

     Explained by: Social support  Explained by: Healthy life expectancy  \
0                           1.106                                 0.741
1                           1.108                                 0.763
2                           1.079                                 0.816
3                           1.172                                 0.772
4                           1.079                                 0.753
..                            ...                                   ...
144                         0.731                                 0.007
145                         0.724                                 0.340
146                         0.202                                 0.407
147                         0.649                                 0.243
148                         0.000                                 0.126

     Explained by: Freedom to make life choices  Explained by: Generosity  \
0                                         0.691                     0.124
1                                         0.686                     0.208
2                                         0.653                     0.204
3                                         0.698                     0.293
4                                         0.647                     0.302
..                                          ...                       ...
144                                       0.405                     0.103
145                                       0.539                     0.027
146                                       0.627                     0.227
147                                       0.359                     0.157
148                                       0.000                     0.122

     Explained by: Perceptions of corruption  Dystopia + residual
0                                      0.481               3.253
1                                      0.485               2.868
2                                      0.413               2.839
3                                      0.170               2.967
4                                      0.384               2.798
..                                       ...                 ...
144                                    0.015               1.800
145                                    0.088               0.648
146                                    0.493               1.095
147                                    0.075               1.205
148                                    0.010               1.895

[149 rows x 20 columns]>
```

```
df.sample
```

```
<bound method NDFrame.sample of          country            Regional  Ladder score  \
0        Finland      Western Europe         7.842
1        Denmark      Western Europe         7.620
2    Switzerland      Western Europe         7.571
3        Iceland      Western Europe         7.554
4    Netherlands      Western Europe         7.464
..           ...                 ...           ...
144      Lesotho  Sub-Saharan Africa         3.512
145     Botswana  Sub-Saharan Africa         3.467
146       Rwanda  Sub-Saharan Africa         3.415
147     Zimbabwe  Sub-Saharan Africa         3.145
148  Afghanistan          South Asia         2.523
```

```
     Standard error of ladder score  upperwhisker  lowerwhisker  \
0                             0.032         7.904         7.780
1                             0.035         7.687         7.552
2                             0.036         7.643         7.500
3                             0.059         7.670         7.438
4                             0.027         7.518         7.410
..                              ...           ...           ...
144                           0.120         3.748         3.276
145                           0.074         3.611         3.322
146                           0.068         3.548         3.282
147                           0.058         3.259         3.030
148                           0.038         2.596         2.449

     Logged GDP per capita  Social support  Healthy life expectancy  \
0                   10.775           0.954                   72.000
1                   10.933           0.954                   72.700
2                   11.117           0.942                   74.400
3                   10.878           0.983                   73.000
4                   10.932           0.942                   72.400
..                     ...             ...                      ...
144                  7.926           0.787                   48.700
145                  9.782           0.784                   59.269
146                  7.676           0.552                   61.400
147                  7.943           0.750                   56.201
148                  7.695           0.463                   52.493

     Freedom to make life choices  Generosity  Perceptions of corruption  \
0                           0.949      -0.098                      0.186
1                           0.946       0.030                      0.179
2                           0.919       0.025                      0.292
3                           0.955       0.160                      0.673
4                           0.913       0.175                      0.338
..                            ...         ...                        ...
144                         0.715      -0.131                      0.915
145                         0.824      -0.246                      0.801
146                         0.897       0.061                      0.167
147                         0.677      -0.047                      0.821
148                         0.382      -0.102                      0.924

     Ladder score in Dystopia  Explained by: Log GDP per capita  \
0                        2.43                             1.446
1                        2.43                             1.502
2                        2.43                             1.566
3                        2.43                             1.482
```

CHECKING COLUMN DATA TYPES

```
df.dtypes
```

```
country                                          str
Regional                                         str
Ladder score                                 float64
Standard error of ladder score               float64
upperwhisker                                 float64
lowerwhisker                                 float64
Logged GDP per capita                        float64
Social support                               float64
Healthy life expectancy                      float64
Freedom to make life choices                 float64
Generosity                                   float64
Perceptions of corruption                    float64
Ladder score in Dystopia                     float64
Explained by: Log GDP per capita             float64
Explained by: Social support                 float64
Explained by: Healthy life expectancy        float64
Explained by: Freedom to make life choices   float64
Explained by: Generosity                     float64
Explained by: Perceptions of corruption      float64
Dystopia + residual                          float64
dtype: object
```

CHECK COLUMNS SHAPE

```
df.shape   # display no. of row and columns
```

```
(149, 20)
```

CHECKING COLUMN NAMES

```
df.columns # display columns names
```

```
Index(['country', 'Regional', 'Ladder score', 'Standard error of ladder score',
       'upperwhisker', 'lowerwhisker', 'Logged GDP per capita',
       'Social support', 'Healthy life expectancy',
       'Freedom to make life choices', 'Generosity',
       'Perceptions of corruption', 'Ladder score in Dystopia',
       'Explained by: Log GDP per capita', 'Explained by: Social support',
       'Explained by: Healthy life expectancy',
       'Explained by: Freedom to make life choices',
       'Explained by: Generosity', 'Explained by: Perceptions of corruption',
       'Dystopia + residual'],
      dtype='str')
```

CHECK INFORMATION

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 149 entries, 0 to 148
Data columns (total 20 columns):
 #   Column                                    Non-Null Count  Dtype
---  ------                                    --------------  -----
 0   country                                   149 non-null    str
 1   Regional                                  149 non-null    str
 2   Ladder score                              149 non-null    float64
 3   Standard error of ladder score            149 non-null    float64
 4   upperwhisker                              149 non-null    float64
 5   lowerwhisker                              149 non-null    float64
 6   Logged GDP per capita                     149 non-null    float64
 7   Social support                            149 non-null    float64
 8   Healthy life expectancy                   149 non-null    float64
 9   Freedom to make life choices              149 non-null    float64
 10  Generosity                                149 non-null    float64
 11  Perceptions of corruption                 149 non-null    float64
 12  Ladder score in Dystopia                  149 non-null    float64
 13  Explained by: Log GDP per capita          149 non-null    float64
 14  Explained by: Social support              149 non-null    float64
 15  Explained by: Healthy life expectancy     149 non-null    float64
 16  Explained by: Freedom to make life choices 149 non-null   float64
 17  Explained by: Generosity                  149 non-null    float64
 18  Explained by: Perceptions of corruption   149 non-null    float64
 19  Dystopia + residual                       149 non-null    float64
dtypes: float64(18), str(2)
memory usage: 23.4 KB
```

Overview of Dataset (World Happiness Report 2021)

Contains data for 149 countries

Measures happiness using Ladder Score (0–10 scale)

Countries are grouped by regions

DROPPING COLUMNS

```
df.drop(columns=['Social support'],inplace= True)
```

```
df
```

	country	Regional	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
0	Finland	Western Europe	7.842	0.032	7.904	7.780	10.775	72.000	0.949	-0.098	0.186
1	Denmark	Western Europe	7.620	0.035	7.687	7.552	10.933	72.700	0.946	0.030	0.179
2	Switzerland	Western Europe	7.571	0.036	7.643	7.500	11.117	74.400	0.919	0.025	0.292
3	Iceland	Western Europe	7.554	0.059	7.670	7.438	10.878	73.000	0.955	0.160	0.673
4	Netherlands	Western Europe	7.464	0.027	7.518	7.410	10.932	72.400	0.913	0.175	0.338
...	...	...	...	...	...	...	...	...	...	...	..
144	Lesotho	Sub-Saharan Africa	3.512	0.120	3.748	3.276	7.926	48.700	0.715	-0.131	0.915
145	Botswana	Sub-Saharan Africa	3.467	0.074	3.611	3.322	9.782	59.269	0.824	-0.246	0.801
146	Rwanda	Sub-Saharan Africa	3.415	0.068	3.548	3.282	7.676	61.400	0.897	0.061	0.167
147	Zimbabwe	Sub-Saharan Africa	3.145	0.058	3.259	3.030	7.943	56.201	0.677	-0.047	0.821
148	Afghanistan	South Asia	2.523	0.038	2.596	2.449	7.695	52.493	0.382	-0.102	0.924

HANDING MISSING VALUES

```
df.isnull().sum()
```

```
country                                   0
Regional                                  0
Ladder score                              0
Standard error of ladder score            0
upperwhisker                              0
lowerwhisker                              0
Logged GDP per capita                     0
Healthy life expectancy                   0
Freedom to make life choices              0
Generosity                                0
Perceptions of corruption                 0
Ladder score in Dystopia                  0
Explained by: Log GDP per capita          0
Explained by: Social support              0
Explained by: Healthy life expectancy     0
Explained by: Freedom to make life choices  0
Explained by: Generosity                  0
Explained by: Perceptions of corruption   0
Dystopia + residual                       0
dtype: int64
```

- THERE IS NO MISSING VALUES

```
df.isnull().median()
```

```
country                                   0.0
Regional                                  0.0
Ladder score                              0.0
Standard error of ladder score            0.0
upperwhisker                              0.0
lowerwhisker                              0.0
Logged GDP per capita                     0.0
Healthy life expectancy                   0.0
Freedom to make life choices              0.0
Generosity                                0.0
Perceptions of corruption                 0.0
Ladder score in Dystopia                  0.0
Explained by: Log GDP per capita          0.0
```

```
Explained by: Social support                  0.0
Explained by: Healthy life expectancy         0.0
Explained by: Freedom to make life choices    0.0
Explained by: Generosity                      0.0
Explained by: Perceptions of corruption       0.0
Dystopia + residual                           0.0
dtype: float64
```

```
df
```

	country	Regional	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
0	Finland	Western Europe	7.842	0.032	7.904	7.780	10.775	72.000	0.949	-0.098	0.186
1	Denmark	Western Europe	7.620	0.035	7.687	7.552	10.933	72.700	0.946	0.030	0.179
2	Switzerland	Western Europe	7.571	0.036	7.643	7.500	11.117	74.400	0.919	0.025	0.292
3	Iceland	Western Europe	7.554	0.059	7.670	7.438	10.878	73.000	0.955	0.160	0.673
4	Netherlands	Western Europe	7.464	0.027	7.518	7.410	10.932	72.400	0.913	0.175	0.338
...	...	...	...	...	...	...	...	...	...	...	..
144	Lesotho	Sub-Saharan Africa	3.512	0.120	3.748	3.276	7.926	48.700	0.715	-0.131	0.915
145	Botswana	Sub-Saharan Africa	3.467	0.074	3.611	3.322	9.782	59.269	0.824	-0.246	0.801
146	Rwanda	Sub-Saharan Africa	3.415	0.068	3.548	3.282	7.676	61.400	0.897	0.061	0.167
147	Zimbabwe	Sub-Saharan Africa	3.145	0.058	3.259	3.030	7.943	56.201	0.677	-0.047	0.821
148	Afghanistan	South Asia	2.523	0.038	2.596	2.449	7.695	52.493	0.382	-0.102	0.924

DAY4

HANDING DUPLICATON

```
df.duplicated().sum()
```

```
np.int64(0)
```

VERIFY THE DATA TYPE

```
df.sample(10)
```

	country	Regional	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Healthy life expectancy	Freedom to make life choices	Generosity	Percepti corrupt
109	Ukraine	Commonwealth of Independent States	4.875	0.052	4.977	4.773	9.436	64.902	0.724	-0.011	0.
79	Libya	Middle East and North Africa	5.410	0.076	5.558	5.262	9.622	62.300	0.771	-0.087	0.
19	Belgium	Western Europe	6.834	0.034	6.901	6.767	10.823	72.199	0.783	-0.153	0.
83	China	East Asia	5.339	0.029	5.397	5.281	9.673	69.593	0.904	-0.146	0.
62	Peru	Latin America and Caribbean	5.840	0.075	5.988	5.692	9.458	68.250	0.822	-0.154	0.
26	Spain	Western Europe	6.491	0.042	6.574	6.408	10.571	74.700	0.761	-0.081	0.
107	Georgia	Commonwealth of Independent States	4.891	0.054	4.998	4.785	9.585	64.300	0.783	-0.238	0.
56	Argentina	Latin America and Caribbean	5.929	0.056	6.040	5.819	9.962	69.000	0.828	-0.182	0.
55	Japan	East Asia	5.940	0.040	6.020	5.861	10.611	75.100	0.796	-0.258	0.
		Middle East									

```
df.shape
```

```
(149, 19)
```

```
df.info
```

```
<bound method DataFrame.info of          country            Regional  Ladder score  \
0        Finland      Western Europe         7.842
1        Denmark      Western Europe         7.620
2    Switzerland      Western Europe         7.571
3        Iceland      Western Europe         7.554
4    Netherlands      Western Europe         7.464
..           ...                 ...           ...
144      Lesotho  Sub-Saharan Africa         3.512
145     Botswana  Sub-Saharan Africa         3.467
146       Rwanda  Sub-Saharan Africa         3.415
147     Zimbabwe  Sub-Saharan Africa         3.145
148  Afghanistan          South Asia         2.523

     Standard error of ladder score  upperwhisker  lowerwhisker  \
0                             0.032         7.904         7.780
1                             0.035         7.687         7.552
2                             0.036         7.643         7.500
3                             0.059         7.670         7.438
4                             0.027         7.518         7.410
..                              ...           ...           ...
144                           0.120         3.748         3.276
145                           0.074         3.611         3.322
146                           0.068         3.548         3.282
147                           0.058         3.259         3.030
148                           0.038         2.596         2.449

     Logged GDP per capita  Healthy life expectancy  \
0                   10.775                   72.000
1                   10.933                   72.700
2                   11.117                   74.400
3                   10.878                   73.000
4                   10.932                   72.400
..                     ...                      ...
144                  7.926                   48.700
145                  9.782                   59.269
146                  7.676                   61.400
147                  7.943                   56.201
148                  7.695                   52.493

     Freedom to make life choices  Generosity  Perceptions of corruption  \
0                           0.949      -0.098                      0.186
1                           0.946       0.030                      0.179
2                           0.919       0.025                      0.292
3                           0.955       0.160                      0.673
4                           0.913       0.175                      0.338
..                            ...         ...                        ...
```

```
144                        0.715      -0.131                     0.915
145                        0.824      -0.246                     0.801
146                        0.897       0.061                     0.167
147                        0.677      -0.047                     0.821
148                        0.382      -0.102                     0.924

     Ladder score in Dystopia  Explained by: Log GDP per capita  \
0                        2.43                             1.446
1                        2.43                             1.502
2                        2.43                             1.566
3                        2.43                             1.482
4                        2.43                             1.501
```

```
df.columns
```

```
Index(['country', 'Regional', 'Ladder score', 'Standard error of ladder score',
       'upperwhisker', 'lowerwhisker', 'Logged GDP per capita',
       'Healthy life expectancy', 'Freedom to make life choices', 'Generosity',
       'Perceptions of corruption', 'Ladder score in Dystopia',
       'Explained by: Log GDP per capita', 'Explained by: Social support',
       'Explained by: Healthy life expectancy',
       'Explained by: Freedom to make life choices',
       'Explained by: Generosity', 'Explained by: Perceptions of corruption',
       'Dystopia + residual'],
      dtype='str')
```

SAVING THE DATA

```
df.to_csv('happences_cleaned.csv',index=False)
```

DAY 3

DATA CLEANING

```
df_new=pd.read_csv("happences_cleaned.csv")
```

```
df_new.head()
```

	country	Regional	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
0	Finland	Western Europe	7.842	0.032	7.904	7.780	10.775	72.0	0.949	-0.098	0.186
1	Denmark	Western Europe	7.620	0.035	7.687	7.552	10.933	72.7	0.946	0.030	0.179
2	Switzerland	Western Europe	7.571	0.036	7.643	7.500	11.117	74.4	0.919	0.025	0.292
3	Iceland	Western Europe	7.554	0.059	7.670	7.438	10.878	73.0	0.955	0.160	0.673
4	Netherlands	Western	7.464	0.027	7.518	7.410	10.932	72.4	0.913	0.175	0.338

SUMMARY OF SATISTATICS

```
df.columns
```

```
Index(['country', 'Regional', 'Ladder score', 'Standard error of ladder score',
       'upperwhisker', 'lowerwhisker', 'Logged GDP per capita',
       'Healthy life expectancy', 'Freedom to make life choices', 'Generosity',
       'Perceptions of corruption', 'Ladder score in Dystopia',
       'Explained by: Log GDP per capita', 'Explained by: Social support',
       'Explained by: Healthy life expectancy',
       'Explained by: Freedom to make life choices',
       'Explained by: Generosity', 'Explained by: Perceptions of corruption',
       'Dystopia + residual'],
      dtype='str')
```

```
df_new.select_dtypes(include='object').mode()
```

```
C:\Users\IC\AppData\Local\Temp\ipykernel_6352\741727298.py:1: Pandas4Warning: For backward compatibility, 'str' dtypes are i
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to
  df_new.select_dtypes(include='object').mode()
```

	country	Regional
0	Afghanistan	Sub-Saharan Africa
1	Albania	NaN
2	Algeria	NaN
3	Argentina	NaN
4	Armenia	NaN
...	...	...
144	Venezuela	NaN
145	Vietnam	NaN
146	Yemen	NaN
147	Zambia	NaN
148	Zimbabwe	NaN

149 rows × 2 columns

```
df_new.select_dtypes(include='number').mean()
```

```
Ladder score                                    5.532839
Standard error of ladder score                  0.058752
upperwhisker                                    5.648007
lowerwhisker                                    5.417631
Logged GDP per capita                           9.432208
Healthy life expectancy                        64.992799
Freedom to make life choices                    0.791597
Generosity                                     -0.015134
Perceptions of corruption                       0.727450
Ladder score in Dystopia                        2.430000
Explained by: Log GDP per capita                0.977161
Explained by: Social support                    0.793315
Explained by: Healthy life expectancy           0.520161
Explained by: Freedom to make life choices      0.498711
Explained by: Generosity                        0.178047
Explained by: Perceptions of corruption         0.135141
Dystopia + residual                             2.430329
dtype: float64
```

```
df_new.select_dtypes(include='number').median()
```

```
Ladder score                                    5.534
Standard error of ladder score                  0.054
upperwhisker                                    5.625
lowerwhisker                                    5.413
Logged GDP per capita                           9.569
Healthy life expectancy                        66.603
Freedom to make life choices                    0.804
Generosity                                     -0.036
Perceptions of corruption                       0.781
Ladder score in Dystopia                        2.430
Explained by: Log GDP per capita                1.025
Explained by: Social support                    0.832
Explained by: Healthy life expectancy           0.571
Explained by: Freedom to make life choices      0.514
Explained by: Generosity                        0.164
Explained by: Perceptions of corruption         0.101
Dystopia + residual                             2.509
dtype: float64
```

```
df.describe()
```

	Ladder score	Standard error of ladder score	upperwhisker	lowerwhisker	Logged GDP per capita	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption	Ladd score Dystop
count	149.000000	149.000000	149.000000	149.000000	149.000000	149.000000	149.000000	149.000000	149.000000	149.
mean	5.532839	0.058752	5.648007	5.417631	9.432208	64.992799	0.791597	-0.015134	0.727450	2.
std	1.073924	0.022001	1.054330	1.094879	1.158601	6.762043	0.113332	0.150657	0.179226	0.
min	2.523000	0.026000	2.596000	2.449000	6.635000	48.478000	0.382000	-0.288000	0.082000	2.
25%	4.852000	0.043000	4.991000	4.706000	8.541000	59.802000	0.718000	-0.126000	0.667000	2.
50%	5.534000	0.054000	5.625000	5.413000	9.569000	66.603000	0.804000	-0.036000	0.781000	2.
75%	6.255000	0.070000	6.344000	6.128000	10.421000	69.600000	0.877000	0.079000	0.845000	2.

```
df.drop(columns=['upperwhisker'],inplace=True)
```

```
df.describe(include=object)
```

```
C:\Users\IC\AppData\Local\Temp\ipykernel_6352\3998025828.py:1: Pandas4Warning: For backward compatibility, 'str' dtypes are
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to
  df.describe(include=object)
```

	country	Regional
count	149	149
unique	149	10
top	Finland	Sub-Saharan Africa
freq	1	36

STATISTICAL SUMMARY OF INSIGHTS

- Top Happiest Countries

Countries like Finland, Denmark, and Switzerland have the highest Ladder scores, meaning people report higher life satisfaction.

- Strong Role of Income (GDP)

Higher GDP per capita is strongly linked with higher happiness levels. Wealthier countries tend to rank higher.

- Social Support Matters

Countries with strong social support systems (family, community, welfare) show higher happiness scores.

- Health is Important

Healthy life expectancy is positively related to happiness. People in healthier countries feel more satisfied.

- Freedom Increases Happiness

The ability to make life choices (freedom) significantly improves well-being.

- Corruption Reduces Happiness

Higher perceptions of corruption are associated with lower happiness levels.

- Generosity Has Smaller Impact

While generosity contributes to happiness, its effect is smaller compared to GDP, health, and social support.

- Regional Differences

Western European countries dominate the top rankings, while lower-ranked countries often have weaker economic and social conditions.

- Dystopia Benchmark

The “Dystopia + residual” value shows that even after accounting for known factors, some variation in happiness remains unexplained.

DAY 5

DATA VISUALIZATION

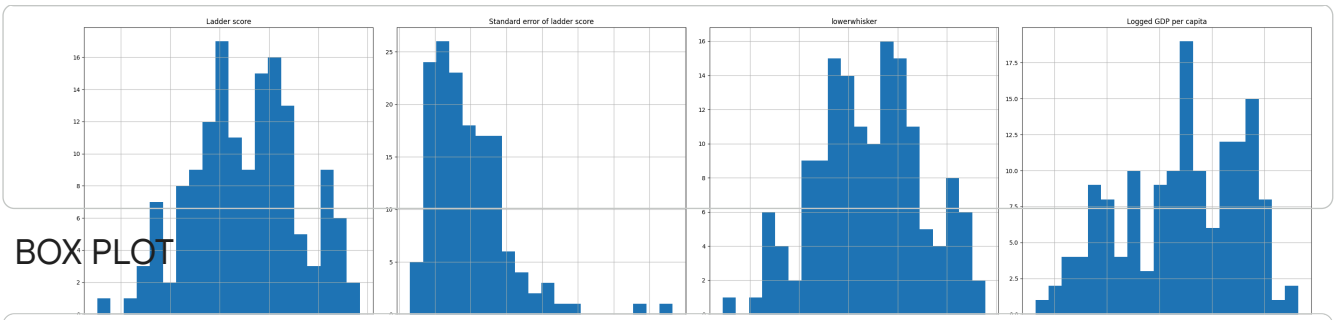
UNIVARITE ANALIYSIS

HISTOGRAM

```
import seaborn as sns
```

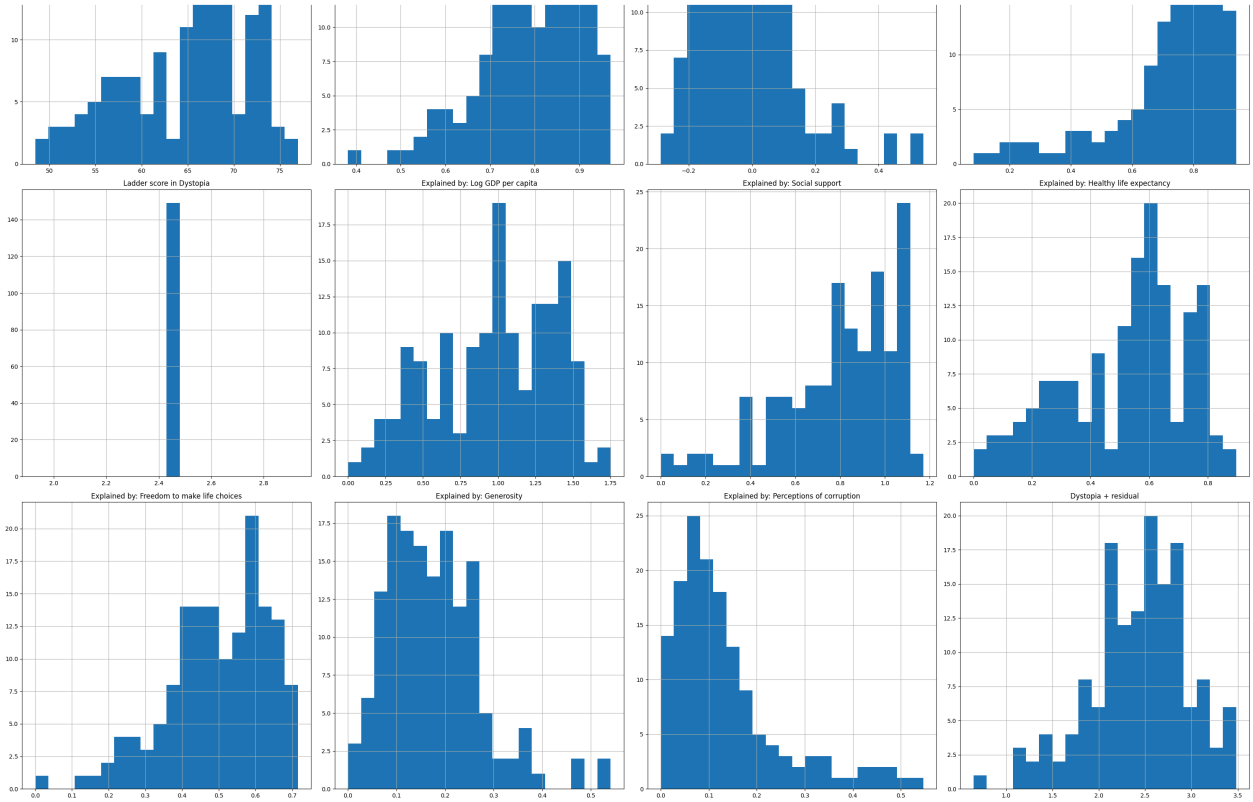
```
import matplotlib.pylab as plt
```

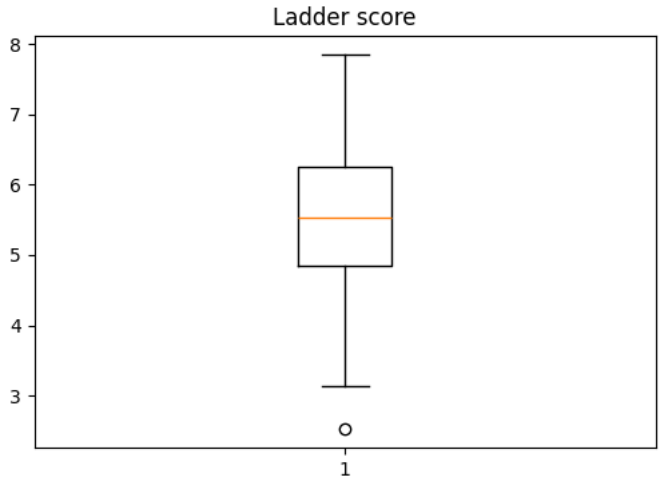
```
df_num = df.copy()
df_num.drop(columns=['Regional','country'],inplace=True)
df_num.hist(figsize=(30,30), bins=20)
plt.tight_layout()
plt.show()
```

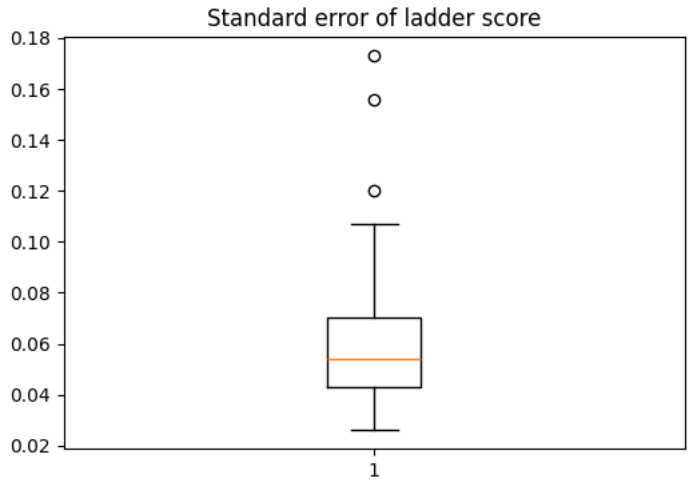


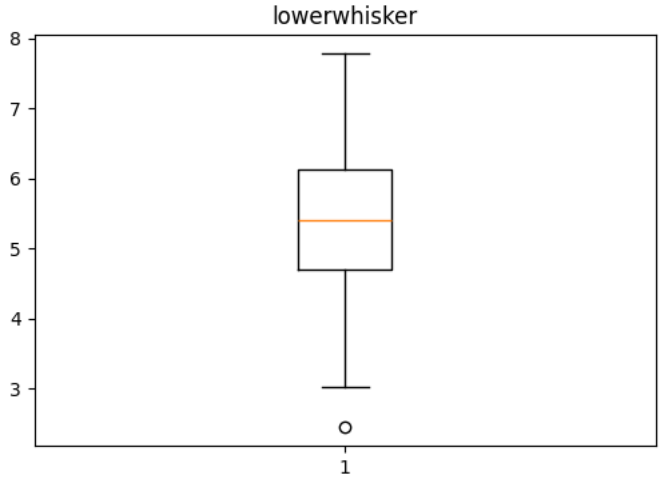
BOX PLOT

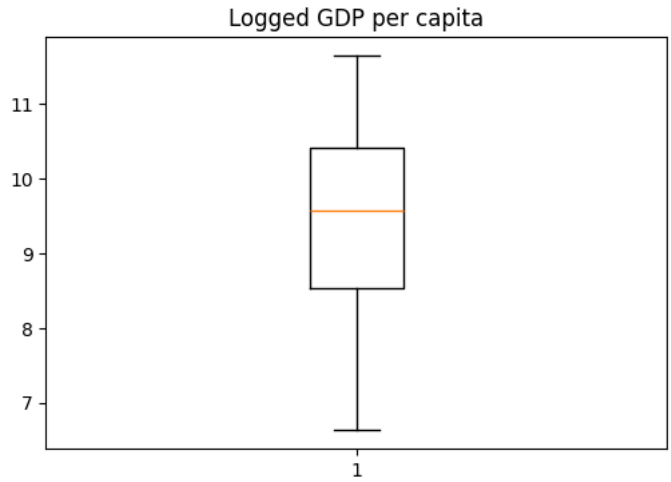
```
for col in df_num.columns:
    plt.figure(figsize=(6,4))
    plt.boxplot(df_num[col])
    plt.title(col)
    plt.show()
```

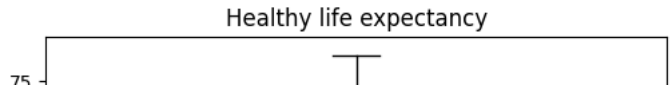



Ladder score
8
7
6
5
4
3
1


Standard error of ladder score
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
1


lowerwhisker
8
7
6
5
4
3
1


Logged GDP per capita
11
10
9
8
7
1


Healthy life expectancy